DFT study report: test calculations before the photocatalyst project

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Purpose. The photocatalyst project computes the electronic structure of 16 organic photocatalysts with DFT. Before working with its pipeline, I ran a series of small test calculations. Each one is on a molecule whose answer is known, so I could understand what each step does and check that the method behaves as expected. Every calculation uses the project's level of theory: **B3LYP/def2-SVP in PySCF**, with ddCOSMO water where stated. Lessons 3–5 are in the exercise notebook `dft_exercises_L3-5.ipynb`, where each result is checked against a reference value.

	Test	Question	Status
1	First DFT calculation	What do the basis set and functional control?	**Reference values — to rerun**
2	Geometry optimisation	What do "optimised" and "converged" mean?	**Reference values — to rerun**
3	Orbitals, IE and TDDFT	What do HOMO/LUMO energies mean, and not mean?	**Completed**
4	SMILES → DFT	How does a string become a computable molecule?	**Completed**
5	Charge and solvent	Why compute eosin Y as −2, and in water?	**Reference values — to rerun**
6	Reproduce the project	Can I reproduce a project result myself?	**Reference values — to rerun**

1 Test 1 — First DFT calculation

Status: **Reference values — to rerun** **Molecule(s):** water

What I ran. Single points at the experimental geometry with three basis sets (STO-3G, def2-SVP, def2-TZVP) and three methods (HF, PBE, B3LYP).

Result. Total energy fell by 1.1 Ha (about 3000 kJ/mol) from STO-3G to def2-TZVP, and changed by about 0.4 Ha between HF and B3LYP. The HOMO energy was −13.8 eV with HF but −8.6 eV with B3LYP (def2-TZVP). The experimental ionisation energy is 12.6 eV.

What I learned. A total energy has no meaning on its own. Only energy differences at the *same* level of theory are meaningful. Orbital energies depend strongly on the functional.

Why this matters for the project

Every number in the project must be labelled with its level of theory and compared only with numbers computed the same way. This is why all 16 molecules use one functional, one basis set and one solvent.

2 Test 2 — Geometry optimisation

Status: Reference values — to rerun **Molecule(s):** water, benzene

What I ran. DFT optimisations starting from force-field structures.

Result. Water: O–H 0.967 Å, H–O–H 103.1° (experiment 0.958 Å, 104.5°). Benzene: C–C 1.398 Å (experiment 1.397 Å), converged in 4 cycles from the MMFF start.

What I learned. An optimisation alternates between an SCF (energy and forces) and a step downhill. "Converged" means both the forces and the step size are below thresholds. A good starting geometry saves expensive DFT steps.

Why this matters for the project

The project starts every molecule from a force-field (MMFF) geometry. For rigid molecules this is already close to DFT. For the flexible and charged dyes (eosin Y, rhodamine B, riboflavin) it was 0.17–0.40 Å off, and eosin Y's very flat twist needed relaxed convergence criteria.

3 Test 3 — Orbitals, ionisation energy and TDDFT

Status: Completed **Molecule(s):** benzene

What I ran. (a) A B3LYP single point and its orbital energies. (b) The ionisation energy three ways. (c) Orbital pictures with `cubegen`. (d) TDDFT for the lowest three excited states.

Results.

Ionisation energy (eV)		Excitations (eV)	
Experiment	9.24	HOMO–LUMO gap	6.72
−HOMO, B3LYP	6.97	TDDFT S_1 ($f = 0.000$)	5.48
−HOMO, HF	9.17	TDDFT S_3 ($f = 0.589$)	7.28
ΔSCF, B3LYP	**9.24**	Experiment: weak / strong band	4.9 / ∼6.9

HOMO/HOMO−1 (−6.97 eV) and LUMO/LUMO+1 (−0.25 eV) are degenerate pairs.

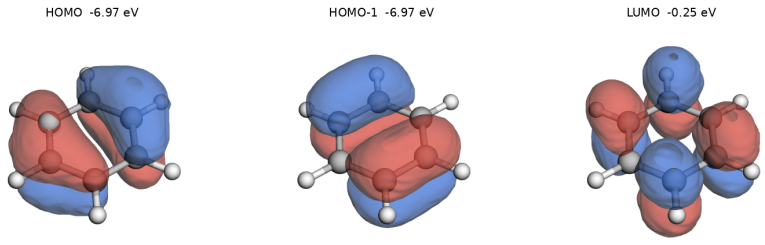


Figure 1: Benzene HOMO, HOMO−1, LUMO (±0.03 a.u.).

What I learned.

- **Self-interaction error.** B3LYP's −HOMO underestimates the ionisation energy by 2.3 eV, because each electron partly repels itself. This pushes occupied orbital energies up. ΔSCF, which subtracts the total energies of the cation and the neutral molecule, cancels most of this error and matches experiment.
- **Three different gaps.** The KS gap (6.72 eV) is an orbital difference. The fundamental gap (ionisation energy minus electron affinity, ≈10.4 eV) needs the ions. The optical gap (4.9 eV) is the lowest absorption, and it is smallest because the excited electron and the hole still attract each other.

- **TDDFT** computes real excitations. Each comes with an oscillator strength f: $f = 0$ means a dark (symmetry-forbidden) state. That is why benzene's lowest band is so weak. S_1 is three times closer to experiment than the KS gap.
- `cubegen` only converts orbital coefficients into a 3D grid for viewing. It changes no results.
- Members of a degenerate pair must not be interpreted alone.

Why this matters for the project

The project's energy-level diagram uses B3LYP HOMO/LUMO energies, so it can show trends between dyes but not real ionisation energies or redox potentials. That is why the water-splitting reference lines were removed from it. The project's comparison with experiment ($r = 0.90$, MAE 0.45 eV) uses the KS gap as a stand-in for absorption; **the project does not use TDDFT yet**, and this test shows it is needed. In Mes-Acr$^+$ the three highest occupied orbitals lie within 13 meV of each other, so the warning about degenerate orbitals applies there too.

4 Test 4 — From SMILES to a DFT calculation

Status: Completed **Molecule(s):** benzene, anthraquinone

What I ran. I wrote my own `smiles_to_atoms` function (SMILES → add H → ETKDG 3D embedding → MMFF → list of atoms and coordinates), then ran DFT single points in vacuum and in water.

```
m = Chem.MolFromSmiles(smiles); q = Chem.GetFormalCharge(m)
mh = Chem.AddHs(m); AllChem.EmbedMolecule(mh, params); AllChem.MMFFOptimizeMolecule(mh)
atoms = [(a.GetSymbol(), tuple(pos[i])) for i, a in enumerate(mh.GetAtoms())]
mol = gto.M(atom=atoms, basis="def2-svp", charge=q, spin=0)
```

Results. Benzene's MMFF geometry gave C–C 1.395 Å, and a HOMO only 0.008 eV different from the DFT-optimised geometry. Anthraquinone:

Geometry	Solvent	HOMO (eV)	LUMO (eV)	Gap (eV)
MMFF (mine)	vacuum	−7.170	−2.976	4.194
MMFF (mine)	water	−7.288	−3.022	4.266
DFT-optimised (project)	water	−7.314	−3.041	4.273

What I learned.

- `MolFromSmiles` returns a *graph* (atoms, bonds, charges) with no coordinates and only implicit hydrogens. A broken SMILES returns `None` without an error.
- DFT receives only nuclei, coordinates, charge and spin, plus the method choices. It never sees bonds or SMILES.
- For rigid anthraquinone the geometry difference (MMFF vs DFT) shifts the gap by only 0.007 eV, and the solvent shifts the orbitals by only about 0.1 eV, because the molecule is neutral with almost no dipole.

Why this matters for the project

This is exactly what the project's Step 1 does (plus conformer search and checks), so I can now read that script. It also explains why MMFF geometries are acceptable for rigid neutral dyes but not automatically for the flexible or charged ones.

5 Test 5 — Charge and solvent

Status: Reference values — to rerun **Molecule(s):** phenol, phenolate

What I ran. Phenol and phenolate ($C_6H_5O^-$) in vacuum and in ddCOSMO water. I also built phenolate once with the charge deliberately left out.

Results.

	HOMO vacuum (eV)	HOMO water (eV)	Solvent shift (eV)
phenol (neutral)	−6.18	−6.26	−0.08
phenolate (−1)	+0.32	−4.15	−4.47

Without the charge, PySCF refused: 49 electrons cannot all be paired.

What I learned. In vacuum the anion's extra electron is not bound (positive HOMO). Water stabilises the ion by 4.5 eV but the neutral molecule by only 0.08 eV, because the solvent responds to charge. For a *dianion*, losing the charge removes two electrons, the count stays even, and no error appears.

Why this matters for the project

Eosin Y, rose bengal and fluorescein are dianions. They must be computed with charge −2 and in solvent, or their orbitals are meaningless. The same silent failure happens with the project's own file: reading `eosinY.xyz` directly into PySCF gives charge 0 and 306 electrons instead of 308, with no warning. That is why the project stores the charge in every xyz file and passes it explicitly. Using water for *every* molecule also keeps all 16 computed at the same level of theory, so they can be compared fairly.

6 Test 6 — Reproducing a project result

Status: Reference values — to rerun **Molecule(s):** anthraquinone

What I ran. A single point with my own code on the project's DFT-optimised anthraquinone geometry, in water.

Result. HOMO −7.3138, LUMO −3.0409 eV: identical to the project's `frontier_orbitals.csv` to four decimal places.

Why this matters for the project

The project's Step 3 is this calculation in a loop over 16 molecules. Reproducing one row independently is the basic check that the pipeline does what I think it does.

7 Mistakes I made and what they taught me

- **Units.** My first ΔSCF ionisation energy for benzene was 0.34. The value was correct but in Hartree: $0.34 \times 27.211 = 9.24$ eV. PySCF always returns Hartree.
- **Wrong molecule.** I used naphthalene's SMILES for anthraquinone. Everything ran; the giveaway was 68 electrons instead of 108. A wrong SMILES is still a valid SMILES, so results must be checked against an independent fact such as the molecular formula. The project does this for every molecule.

8 Summary and next steps

Main things I learned. Compare only energy differences at the same level of theory. B3LYP orbital energies give trends, not absolute ionisation energies or redox potentials. Absorption needs TDDFT. The charge must be passed explicitly, because losing two charges fails silently. Ions need a solvent model. And every result should be checked against something independent.

Next steps.

1. Rerun Tests 1, 2, 5 and 6 myself.
2. Run TDDFT on anthraquinone and compare with its measured 325 nm band (the first step towards adding TDDFT to the project).
3. Rerun the project's Step 3 on the three newly optimised dyes, then Step 5, to see whether the geometry changes their orbital energies.

Questions for my PI.

- Which functional does the group use for TDDFT on dyes, especially charge-transfer ones?
- Should each molecule be computed in its experimental solvent rather than water?
- How does the group usually compute redox potentials (ΔSCF in solvent, or another approach)?